Toward Dynamic Romance Systems: Integrating Emotion Modeling and Utility AI in Games

Joaquin Arredondo, Camila Yeremin, April Grow
Computer Science and Software Engineering, *California Polytechnic State University*, San Luis Obispo, CA, USA
joaquinanthonyarredondo@gmail.com, yereminsasha@gmail.com, amgrow@calpoly.edu

***Abstract*—Romance mechanics in contemporary video games are frequently implemented through affinity meters and scripted dialogue trees, reducing relationships to linear progression systems. This paper presents the design and Unity-based implementation of a modular romance framework that models interpersonal dynamics using affective state and utility-based decision-making. Our architecture integrates the Pleasure–Arousal–Dominance (PAD) emotional model, personality bias parameters inspired by the NERIS (MBTI-derived) spectrum dimensions, and a Utility AI action-selection system. Player interactions are abstracted into intent categories that produce continuous emotional updates (△PAD), which influence NPC decision-making through response curves and weighted utility scoring. We describe the system architecture and provide a preliminary A–B simulation demonstrating that identical interaction sequences can produce different PAD trajectories and responses under different personality configurations. This work focuses on operationalizing psychologically inspired constructs as tunable state variables for interactive systems, rather than evaluating perceived realism in players.**

***Index Terms*—Game AI, Utility AI, Affective Computing, Emotion Modeling, Social Simulation, Non-Player Characters (NPCs), Interactive Systems**

I. Introduction

Romantic interaction is a recurring feature in life simulations, role-playing games, and narrative-driven titles. In most commercial implementations, however, romance is structured through one-dimensional affinity meters, scripted dialogue trees, and threshold-based progression systems. While these mechanics provide clarity and reward feedback, they model relationships as transactional and/or as linear optimization problems, rather than evolving emotional processes. Relationships become goals to unlock instead of dynamic systems shaped by mood, personality, and context. As a result, many games struggle to represent the ambiguity, fluctuation, and emotional reciprocity that characterize real-world romantic attachment.

Research in computational social systems has explored richer representations of interpersonal dynamics. Architectures such as Comme il Faut (CIF) [1], demonstrated in *Prom Week* [2], model social interaction through structured predicates, traits, and rule-based evaluation of social exchanges. Successor systems such as *Ensemble* [3] expand this approach by representing social state as a network of relational variables governed by authored rules and social physics. Praxis-based and similar expressive AI systems further investigate simulation-driven narrative through formalized representations of character goals and social reasoning [4], [5]. These systems demonstrate that interpersonal behavior can emerge from computational structure rather than scripted branching alone.

However, many expressive social AI systems rely heavily on discrete symbolic predicates and authored rule sets, requiring significant design specification and authoring effort. In contrast, affective computing research offers compact, continuous representations of emotional state, such as the Pleasure–Arousal–Dominance (PAD) model [6]–[8], while modern game AI frequently employs Utility AI frameworks for adaptive action selection [9] [10]. These approaches emphasize scalable, numeric evaluation of state rather than symbolic rule evaluation.

This work is motivated by the need for an alternative framework—one that models romance not as a transactional reward structure, but as a dynamic experience grounded in psychological theory. We argue that player choice should meaningfully influence non-player character (NPC) emotional state, and that these emotional changes should in turn shape future relational development. To support this, we design a modular system centered on affect modeling (PAD), personality structure, and Utility AI decision-making; we also include a relationship module based on Sternberg's Triangular Theory of Love as an extensible component of the architecture. Rather than relying on static thresholds, our system represents NPCs as agents with emotional baselines, personality-driven preferences, and context-sensitive motivations. Player actions influence emotional state variables, which reshape future decision priorities, creating feedback loops rather than fixed romance paths.

The primary goal of this work is architectural: to demonstrate how affect-driven utility systems can support dynamic romance mechanics suitable for interactive game development. We provide implementation details and a preliminary Unity-based A–B simulation illustrating how personality weighting can produce differentiated PAD trajectories and responses under identical player input sequences.

II. Related Work

A. Psychology and Affective Computing

Affective computing formalized the study of systems capable of modeling and responding to emotional state [7]. Dimensional emotion models, particularly the Pleasure–Arousal–Dominance (PAD) framework introduced by Mehrabian and Russell [6], represent affect as three continuous axes: pleasure (valence), arousal (intensity), and dominance (perceived control). Unlike categorical emotion labels, PAD enables gradual, quantifiable shifts in emotional state, making it suitable for real-time interactive systems.

PAD and related dimensional emotion models have been applied in socially interactive robotics to regulate expressive behavior [11], integrated into agent architectures to influence decision-making [8], used in expressive avatars to modulate animation and dialogue intensity [12], and incorporated into interactive game systems for adaptive responses [13]. Collectively, this work supports our use of PAD as a dynamic state variable embedded directly within utility-based action evaluation rather than as a purely representational layer.

Sternberg's Triangular Theory of Love (TOL) [14] provides a structured representation of relationship state through intimacy, passion, and commitment. Unlike a single affection meter, this model supports multiple relational trajectories and "love shapes," enabling romance to evolve along diverse paths rather than a linear progression.

In addition to affect modeling, our framework draws from established psychological theories of personality and romantic attachment to influence how NPCs interpret social cues and prioritize actions. The Five-Factor Model (Big Five) is widely regarded as the dominant empirical framework in personality psychology due to its cross-cultural robustness and strong psychometric support [15], [16]. Although the Myers–Briggs Type Indicator (MBTI) has been criticized for categorical rigidity when used diagnostically [17], its typological structure remains influential and interpretable in applied contexts. We therefore adopt the NERIS Type Explorer® framework, a contemporary extension of MBTI that organizes personality along five continuous spectrums (Energy, Mind, Nature, Tactics, Identity) while retaining familiar type codes for readability [18]. This hybrid trait–type formulation aligns more closely with dimensional trait theory while preserving the clarity of typological labels. In our system, these spectrums function as computational bias weights that modulate emotional sensitivity and utility scoring rather than as psychological assessments; preliminary research has reported internal consistency of the NERIS instrument within sampled populations [19].

B. Interpersonal and Social AI Systems

Computational modeling of social interaction has been explored extensively in expressive AI research. The Comme il Faut (CIF) architecture [1] introduced a rule-based social reasoning framework in which character traits, social predicates, and relationship variables interact through authored rules. In *Prom Week* [2], CIF enabled emergent social scenarios by evaluating the preconditions and effects of social exchanges within a structured social state.

Building on this work, *Ensemble* [3] generalized social modeling through networks of predicates and rule-based transitions, allowing designers to declaratively encode social norms and dispositions. Praxis-based systems and related narrative architectures [4], [5], [20] similarly model character intention and social dynamics through structured reasoning over goals and beliefs. These systems prioritize explainable social logic and narrative coherence.

While highly expressive, such architectures often require substantial authoring effort: designers must explicitly define predicates, rules, preconditions, and effect permutations. Our framework differs in emphasis. Rather than relying primarily on symbolic rule evaluation, we embed interpersonal dynamics within continuous emotional variables and utility-based decision scoring. This approach reduces reliance on large rule sets while still producing differentiated relational behavior.

C. Utility AI and Simulation-Driven Behavior

Utility AI is widely used in modern game development as a method for adaptive decision-making based on weighted considerations and response curves [9], [10]. Popularized in part through analyses of systems such as *The Sims* [21], utility-based architectures allow agents to continuously evaluate internal needs and environmental context when selecting actions. Instead of scripting fixed transitions, actions are scored according to how well they satisfy current conditions, producing adaptive behavior as internal variables fluctuate.

Traditionally, Utility AI has been applied to survival metrics such as hunger, energy, and resource management. In this work, we extend the utility framework to interpersonal mechanics by incorporating PAD-based emotional state and personality bias weights into the scoring process. This allows NPC responses to shift as affect changes, without relying on static progression flags or large authored rule sets.

III. System Design

At a high level, our system maps player actions to perceived intent, which produces a change in the NPC's emotional state (ΔPAD). This emotional shift is then filtered through personality (MBTI) and relationship structure (TOL), and finally evaluated through a Utility AI framework to determine the NPC's response. This creates a closed emotional-decision loop in which player behavior alters emotional state, and emotional state reshapes future behavior.

A. Intent-Based interaction

Rather than binding emotional changes directly to individual dialogue options, player actions are categorized into (currently) 9 intent classes: Affection, Desire, Bonding, Trust, Respect, Playfulness, Security, Conflict, and Manipulation. Each intent has 5 actions, with each action carrying an associated intensity value. This hierarchical intent-to-action mapping was inspired by CiF's social move design, where moves were internally labeled and designed as having intended changes to the relationships, such as "Coolness-Up," "Buddy-Down," or "Start-Dating" [1]. Our intents are much more decoupled from the other sub-systems in our framework. Intent classification allows similar emotional effects to emerge across multiple narrative contexts while preserving implementation flexibility.

B. Affective State Representation

NPC emotional state is represented using the PAD model [6]. Pleasure captures valence, arousal represents emotional intensity, and dominance reflects perceived agency or control. These values are continuous and updated incrementally. When a player action is performed, it generates a base emotional adjustment. This adjustment is scaled by action intensity and modified by both personality parameters and current relationship state. The resulting ΔPAD is applied to the NPC's emotional state, enabling gradual evolution rather than abrupt state transitions.

C. Relationship Structure

Relationship status is modeled using three variables derived from Sternberg's Triangular Theory of Love: intimacy, passion, and commitment [14]. These variables evolve independently over time. Emotional state influences how quickly these variables shift, and existing relationship levels modulate how new actions are perceived. For example, high intimacy may amplify bonding-related intents, while low commitment may dampen high-intensity romantic gestures. This approach to relationships is drastically different to those in other research, where characters are typically either in or out of a relationship or have one dimension of attraction. It also allows for nuance in what CiF would call social moves: an agent's action can do more than succeed or fail, but it can partially succeed. This separation allows relationships to develop incrementally along multiple trajectories.

D. Personality Biasing

Personality parameters are implemented using the NERIS Type Explorer® spectrum model, which represents personality along five independent dimensions: Energy (Extraversion–Introversion), Mind (Observant–Intuitive), Nature (Thinking–Feeling), Tactics (Judging–Prospecting), and Identity (Assertive–Turbulent). Rather than treating personality as categorical type labels, these spectrums are encoded as continuous weight multipliers that influence emotional sensitivity and action preference. For example, higher Energy values increase the weighting of socially proactive or affiliative actions, while lower Energy biases toward reserved or reflective responses. Nature influences the relative weighting of emotionally driven versus analytical decisions, and Identity modulates emotional stability by amplifying or dampening the effect of recent emotional deltas (ΔPAD). These spectrums do not function as psychological diagnoses; instead, they serve as computational bias parameters that shape how NPCs interpret emotional changes and prioritize actions. This approach ensures behavioral consistency across interactions without requiring extensive symbolic rule authoring.

E. Utility-Based Action Selection

NPC action selection is driven by a Utility AI framework operating over continuously updated emotional and relational state. When a player action occurs, it is mapped to an intent category associated with a base PAD bias vector (ΔP, ΔA, ΔD) scaled by action intensity. This emotional change is modulated by a relationship amplifier based on current intimacy, passion, and commitment values, then applied to the NPC's PAD state.

From the updated PAD profile, the system derives emotional considerations used in scoring candidate actions. These include state-based conditions (P_High/Low, A_High/Low, D_High/Low) and delta-based triggers (ΔP, ΔA, ΔD) that capture recent emotional shifts, along with an IntentMatch consideration rewarding contextually aligned responses. Each consideration is transformed through a configurable response curve (linear, threshold, soft-gate, or delta-sensitive) to determine its contribution to an action's utility.

Consideration outputs are normalized and aggregated into a final utility score. Personality spectrum weights are then applied multiplicatively to bias selection according to stable trait orientations (e.g., Energy influencing sociability, Identity influencing emotional volatility). The action with the highest weighted utility is executed. Because emotional, relational, and personality variables evolve continuously, action priorities shift dynamically over time, producing non-linear relational behavior without reliance on scripted thresholds.

IV. Implementation

The framework was implemented as a modular simulation prototype in Unity. Each NPC agent is composed of four primary components: (1) an emotional state module storing continuous PAD values, (2) a relationship module maintaining intimacy, passion, and commitment variables, (3) a personality module encoding five normalized spectrum weights, and (4) a Utility AI evaluator responsible for action scoring and selection. All of the emotional and relationship state variables, ΔPAD changes, personality influences, and utility AI curves are implemented as data-driven configuration assets, allowing all parameters to be tuned without modifying core logic. The progression of a player interaction follows the following steps:

1) Player action
2) Intent classification
3) ΔPAD update (base bias $\times$ intensity)
4) Relationship amplification (TOL modulation)
5) Updated PAD state
6) Utility evaluation based on:
    - Current PAD values
    - ΔPAD triggers
    - Relationship variables
    - Personality spectrum weights
7) Select highest-utility action
8) Execute NPC action

V. Preliminary Validation

To evaluate whether personality spectrum weights meaningfully influence emotional trajectories and action selection, we conducted a controlled A–B simulation using two NPC configurations with identical initial PAD states but different personality profiles:

- **NPC A:** ENFJ-A (high Energy, high Nature, Assertive Identity)
- **NPC B:** INFJ-T (low Energy, high Nature, Turbulent Identity)

Both NPCs were subjected to the same three-step interaction sequence:

1) `TeaseHarsh`
2) `Apology`
3) `Promise`

Initial PAD state for both agents was:

$$P = 0.7, \quad A = 0.7, \quad D = 0.7$$

A. Emotional Trajectory and Behavioral Divergence

a) Step 1: TeaseHarsh: Following the conflict-oriented interaction, both NPCs experienced a decrease in pleasure and an increase in arousal, entering an anger-dominant state. Neither NPC chose to acknowledge the player.

- **ENFJ-A:** Selected *no response* (ignore behavior).
  Updated PAD: $P = 0.25,\ A = 0.9,\ D = 0.85$
- **INFJ-T:** Selected *no response* (ignore behavior).
  Updated PAD: $P = 0.35,\ A = 1.0,\ D = 1.0$

Despite identical intent mapping and base ΔPAD values, personality multipliers influenced the severity of their emotional state.

b) Step 2: Apology: The apology interaction produced a positive pleasure shift in both agents, but recovery rates differed:

- **ENFJ-A:** Emotional state partially recovered but remained elevated in arousal; selected *Demand*.
  Updated PAD: $P = 0.45,\ A = 0.81,\ D = 0.66$
- **INFJ-T:** Transitioned to a joy-dominant state after a single apology; selected *Defend*.
  Updated PAD: $P = 0.55,\ A = 0.91,\ D = 0.81$

The Turbulent Identity parameter amplified responsiveness to positive ΔP triggers, accelerating emotional recovery relative to the Assertive configuration.

c) Step 3: Promise: The commitment-oriented interaction further increased pleasure for both agents:

- **ENFJ-A:** Reached a joy-dominant state after the promise. Still selected *Demand*.
  Final PAD: $P = 0.61,\ A = 0.73,\ D = 0.52$
- **INFJ-T:** Maintained a joy-dominant state with higher overall arousal and dominance values. Also selected *Demand*.
  Final PAD: $P = 0.71,\ A = 0.83,\ D = 0.67$

B. Interpretation

This controlled comparison demonstrates that identical player inputs can produce distinct emotional trajectories and some different action selections under different personality configurations, even when starting PAD states are identical. Personality spectrum weights influenced:

- The magnitude of emotional recovery following apology
- The persistence of elevated arousal
- The likelihood of disengagement versus defensive response
- Final dominance levels after reconciliation attempts

Importantly, divergence occurred without modifying intent mappings or base ΔPAD values, indicating that personality multipliers within the Utility AI pipeline materially affect emotional dynamics and behavioral selection.

VI. Discussion

This work explores an alternative to threshold-based romance systems by embedding affective state and personality biasing directly within a utility-driven decision architecture. Rather than treating romance as a linear progression variable, the proposed framework models interaction as a feedback loop between player intent, emotional state, and action selection. Our Unity A–B simulation shows that identical interaction sequences can yield distinct PAD trajectories and, in some cases, different responses under different personality configurations, even when starting from the same PAD state. This suggests that continuous affective modeling combined with weighted utility evaluation can generate context-sensitive interpersonal behavior while maintaining architectural simplicity.

However, the present validation is limited to system-level simulation rather than player-facing evaluation. While emotional divergence and action differentiation were observed computationally, this work does not yet assess perceived realism, immersion, or player interpretation. Additionally, the relationship model was not fully integrated into the validation phase, and its interaction with PAD and personality requires further study.

Future work will focus on formal trajectory analysis across larger simulation batches, integration of relationship-state amplification into validation trials, and user studies measuring perceived relational depth. Additional extensions may include attachment-style modeling, adaptive dialogue generation, and dynamic personality shifts over long-term interaction.

VII. Conclusion

This paper presented the design and Unity-based implementation of a modular romance framework integrating affect modeling (PAD), spectrum-based personality biasing, and Utility AI decision-making. By representing emotion as continuous state and embedding personality weights directly within action evaluation, the system produces adaptive relational behavior without reliance on static affinity meters or large rule-based social engines.

Preliminary A–B simulations demonstrate that personality parameters materially influence emotional recovery, arousal persistence, and action selection under identical player input sequences. These findings support the viability of affect-driven utility architectures for modeling interpersonal mechanics in interactive systems. While further empirical evaluation is needed, the presented framework establishes a scalable computational foundation for dynamic romance mechanics in games.

REFERENCES

[1] J. McCoy, M. Treanor, B. Samuel, N. Wardrip-Fruin, and M. Mateas, "Comme il Faut: A system for authoring playable social models," in *Proc. 7th AAAI Conf. Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, 2011.

[2] J. McCoy, M. Treanor, B. Samuel, A. A. Reed, M. Mateas, and N. Wardrip-Fruin, "Prom Week," in *Proc. AAAI Conf. Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, 2013.

[3] B. Samuel, J. McCoy, M. Mateas, and N. Wardrip-Fruin, "The Ensemble Engine: Next-generation social physics," in *Proc. Foundations of Digital Games (FDG)*, 2015.

[4] B. Samuel, M. Mateas, and N. Wardrip-Fruin, "Praxis: A logic-based social simulation framework," in *Proc. AAAI Conf. Artificial Intelligence and Interactive Digital Entertainment (AIIDE)*, 2015.

[5] R. Evans and E. Short, "Versu—A simulationist storytelling system," *Computer*, vol. 47, no. 2, pp. 72–80, 2014.

[6] A. Mehrabian and J. A. Russell, *An Approach to Environmental Psychology*. Cambridge, MA, USA: MIT Press, 1974.

[7] R. W. Picard, *Affective Computing*. Cambridge, MA, USA: MIT Press, 1997.

[8] J. Gratch and S. Marsella, "A domain-independent framework for modeling emotion," *Cognitive Systems Research*, vol. 5, no. 4, pp. 269–306, 2004.

[9] D. Mark and K. Dill, "Improving AI decision modeling through utility theory," Game Developers Conference (GDC) AI Summit, 2010.

[10] D. Graham, "An Introduction to Utility Theory," in *Game AI Pro: Collected Wisdom of Game AI Professionals*, S. Rabin, Ed. Boca Raton, FL, USA: A K Peters/CRC Press, 2014, pp. 113–126.

[11] C. Breazeal, "Emotion and sociable humanoid robots," *Int. J. Human-Computer Studies*, vol. 59, no. 1–2, pp. 119–155, 2003.

[12] P. Gebhard, "ALMA: A layered model of affect," in *Proc. 4th Int. Joint Conf. Autonomous Agents and Multiagent Systems (AAMAS)*, 2005.

[13] G. N. Yannakakis and J. Togelius, *Artificial Intelligence and Games*. Cham, Switzerland: Springer, 2018.

[14] R. J. Sternberg, "A triangular theory of love," *Psychological Review*, vol. 93, no. 2, pp. 119–135, 1986.

[15] R. R. McCrae and P. T. Costa, "Personality trait structure as a human universal," *American Psychologist*, vol. 52, no. 5, pp. 509–516, 1997.

[16] O. P. John and S. Srivastava, "The Big Five trait taxonomy: History, measurement, and theoretical perspectives," in *Handbook of Personality: Theory and Research*, 2nd ed., L. A. Pervin and O. P. John, Eds. New York, NY, USA: Guilford Press, 1999.

[17] D. J. Pittenger, "Cautionary comments regarding the Myers–Briggs Type Indicator," *Consulting Psychology Journal: Practice and Research*, vol. 57, no. 3, pp. 210–221, 2005.

[18] NERIS Analytics Limited, "Our Theory," *16Personalities*, accessed 2026. [Online]. Available: `https://www.16personalities.com/articles/our-theory`.

[19] K. Makwana and G. B. Dave, "Confirmatory factor analysis of NERIS Type Explorer® scale," *Int. J. Management*, vol. 11, no. 9, 2020.

[20] A. S. Rao and M. P. Georgeff, "BDI agents: From theory to practice," in *Proc. 1st Int. Conf. Multi-Agent Systems (ICMAS)*, 1995.

[21] R. Zubek, "Needs-based AI," 2004.